

Instruction Format Specification for a Two-Pass Assembler Targeting IA-32

32-bit x86 (IA-32) Architecture — Excluding Instruction Prefixes

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1. Introduction

This document specifies the instruction format and encoding rules for a two-pass assembler targeting the 32-bit x86 (IA-32) architecture. It covers opcode encoding, register and memory addressing, ModR/M and SIB bytes, immediates, displacements, relative jumps, and assembler directives.

IA-32 instructions are variable length. Multi-byte values are stored in little-endian format, with the least significant byte stored first.

2. IA-32 INSTRUCTION FORMAT

An IA-32 instruction may contain an opcode, ModR/M byte, SIB byte, displacement, and immediate field. Fields are emitted in this order: **Opcode → ModR/M -> SIB -> Displacement -> Immediate.**

The detailed visual layout is provided in the illustrated reference sheet below.

Figure 2-1. IA-32 Architecture Instruction Format

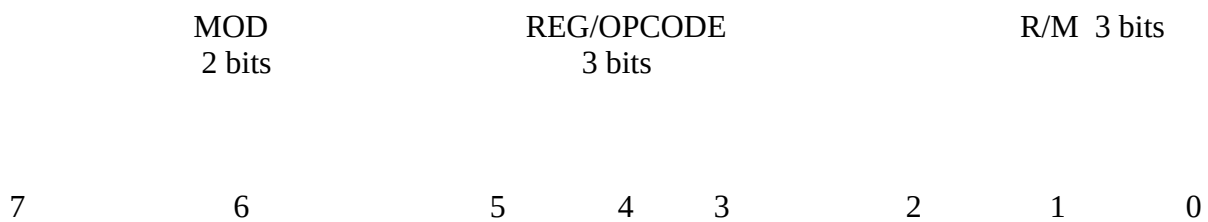
Opcode	ModR/M	SIB	Displacement	Immediate
1-, 2-, or 3-byte opcode	1 byte (if required)	1 byte (if required)	Address displacement of 1, 2, or 4 bytes or none ³	Immediate data of 1, 2, or 4 bytes or none ³

Index	Mnemonics	Opcode	No. of Operands	Op1	Op2	Mod r/m	3-5	+ rd
A	ADD	01 /r	2	r/m	r	Yes	r	No
	ADD	03 /r	2	r	r/m	Yes	r	No
	ADD	81/0 id	2	r/m	imm	Yes	000	No
	ADD	05 id	2	EAX	imm	No	-1	No
	AND	21 /r	2	r/m	r	Yes	r	No
	AND	23 /r	2	r	r/m	Yes	r	No
	AND	25 id	2	EAX	imm	No	-1	No
	AND	81 /4 id	2	r/m	imm	Yes	100	No
C	CMP	39 /r	2	r/m	r	Yes	r	No
	CMP	3B /r	2	r	r/m	Yes	r	No
	CMP	3D id	2	EAX	imm	No	-1	No
	CMP	81 /7 id	2	r/m	imm	Yes	111	No
D	DEC	r/m	1	r/m	-	Yes	001	No
	DEC	48 + rd	1	r	-	No	-1	Yes
	DIV	F7 /6	1	r/m	-	Yes	110	No
I	INC	FF /0	1	r/m	-	Yes	000	No
	INC	40 + rd	1	r	-	No	-1	Yes
M	MOV	89 /r	2	r/m	r	Yes	r	No
	MOV	8B /r	2	r	r/m	Yes	r	No
	MOV	C7 /0 id	2	r/m	imm	Yes	000	No
	MOV	B8 + rd id	2	r	imm	No	-1	Yes
	MUL	F7 /4	1	r/m	-	Yes	100	No
O	OR	09 /r	2	r/m	r	Yes	r	No
	OR	0B /r	2	r	r/m	Yes	r	No
	OR	0D id	2	EAX	imm	No	-1	No

	OR	81 /1 id	2	r/m	imm	Yes	001	No
S	SUB	29 /r	2	r/m	r	Yes	r	No
	SUB	2B /r	2	r	r/m	Yes	r	No
	SUB	2D id	2	EAX	imm	No	-1	No
	SUB	81 /5 id	2	r/m	imm	Yes	101	No
X	XOR	31 /r	2	r/m	r	Yes	r	No
	XOR	33 /r	2	r	r/m	Yes	r	No
	XOR	35 id	2	EAX	imm	No	-1	No
	XOR	81 /6 id	2	r/m	imm	Yes	110	No

2.1.3 ModR/M Byte

The ModR/M byte contains three fields:



The ModR/M byte is calculated as:

$$\text{ModR/M} = (\text{MOD} \ll 6) + (\text{REG} \ll 3) + \text{R/M}$$

Binary	32-bit Register
000	EAX
001	ECX
010	EDX
011	EBX
100	ESP
101	EBP
110	ESI

INSTRUCTION TABLE**ADD—Add**

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 3-5) Register code	R/M	+rd	SIB	Immediate/ Displacement
ADD	r/m32, r32	01/r	Required	Variable	Register code	Variable	—	If required	—
ADD	r32, r/m32	03/r	Required	Variable	Register code	Variable	—	If required	—
ADD	r/m32, imm32	81/0 id	Required	Variable	000	Variable	—	If required	id
ADD	r/m32, imm8	83/0 ib	Required	Variable	000	Variable	—	If required	ib(sign-extended)
ADD	EAX, imm32	05 id	No	—	—	—	—	No	id

SUB- Subtract

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 3-5) Register code	R/M	+rd	SIB	Immediate/ Displacement
SUB	r/m32, r32	29 /r	Required	Variable	Register code	Variable	—	If required	—
SUB	r32, r/m32	2B /r	Required	Variable	Register code	Variable	—	If required	—
SUB	r/m32, imm32	81 /5 id	Required	Variable	101	Variable	—	If required	id
SUB	r/m32, imm8	83 /5 ib	Required	Variable	101	Variable	—	If required	ib(sign-extended)
SUB	EAX, imm32	2D id	No	—	—	—	—	No	id

MOV—Move

Instruction	Operands	Opcode	ModR/M	Mod	Reg(bits 3-5)	R/M	+rd	SIB	Immediate/Displacement
MOV	r/m32, r32	89 /r	Required	Variable	Register code	Variable	—	If required	—
MOV	r32, r/m32	8B /r	Required	Variable	Register code	Variable	—	If required	—
MOV	r/m32, imm32	C7 /0 id	Required	Variable	000	Variable	—	If required	id
MOV	r32, imm32	B8+ rd id	No	—	—	—	rd	No	id

CMP—Compare Two Operands

Instruction	Operands	Opcode	ModR/M	Mod	Reg(bits 3-5)	R/M	+rd	SIB	Immediate/Displacement
CMP	r/m32, r32	39 /r	Required	Variable	Register code	Variable	—	If required	—
CMP	r32, r/m32	3B /r	Required	Variable	Register code	Variable	—	If required	—
CMP	r/m32, imm32	81 /7 id	Required	Variable	111	Variable	—	If required	id
CMP	r/m32, imm8	83 /7 ib	Required	Variable	111	Variable	—	If required	ib(sign-extended)
CMP	EAX, imm32	3D id	No	—	—	—	—	No	id

INC—Increment by 1

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 3-5)	R/M	+rd	SIB	Immediate/Displacement
INC	r/m32	FF /0	Required	Variable	000	Variable	—	If required	—
INC	r32	40+ rd	No	—	—	—	rd	No	—

INC +rd

Register	rd	Opcode
EAX	0	40
ECX	1	41
EDX	2	42
EBX	3	43
ESP	4	44
EBP	5	45
ESI	6	46
EDI	7	47

DEC—Decrement by 1

Instruction Opetands		Opcode	ModR/M	Mod	Reg(bits 3-5)	R/M	+rd	SIB	Immediate/Displacement
DEC	r/m32	FF /1	Required	Variable	001	Variable	—	If required	—
DEC	r32	48+rd	No	—	—	—	rd	No	—

DEC +rd

Register	rd	Opcode
EAX	0	48
ECX	1	49
EDX	2	4A
EBX	3	4B
ESP	4	4C
EBP	5	4D
ESI	6	4E
EDI	7	4F

AND—Logical AND

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 5) Register code	3-R/M	+rd	SIB	Immediate/ Displacement
AND	r/m32, r32	21 /r	Required	Variable	Register code	Variable	—	If required	—
AND	r32, r/m32	23 /r	Required	Variable	Register code	Variable	—	If required	—
AND	r/m32, imm32	81 /4 id	Required	Variable	100	Variable	—	If required	id
AND	r/m32, imm8	83 /4 ib	Required	Variable	100	Variable	—	If required	ib(sign- extended)
AND	EAX, imm32	25 id	No	—	—	—	—	No	id

OR—Logical Inclusive OR

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 5) Register code	3-R/M	+rd	SIB	Immediate/ Displacement
OR	r/m32, r32	09 /r	Required	Variable	Register code	Variable	—	If required	—
OR	r32, r/m32	0B /r	Required	Variable	Register code	Variable	—	If required	—
OR	r/m32, imm32	81 /1 id	Required	Variable	001	Variable	—	If required	id
OR	r/m32, imm8	83 /1 ib	Required	Variable	001	Variable	—	If required	ib(sign- extended)
OR	EAX, imm32	0D id	No	—	—	—	—	No	id

XOR—Logical Inclusive XOR

Instruction	Opetands	Opcode	ModR/M	Mod	Reg(bits 5)	3-R/M	+rd	SIB	Immediate/ Displacement
XOR	r/m32, r32	31 /r	Required	Variable	Register code	Variable	—	If required	—
XOR	r32, r/m32	33 /r	Required	Variable	Register code	Variable	—	If required	—
XOR	r/m32, imm32	81 /6 id	Required	Variable	110	Variable	—	If required	id
XOR	r/m32, imm8	83 /6 ib	Required	Variable	110	Variable	—	If required	ib(sign- extended)
XOR	EAX, imm32	35 id	No	—	—	—	—	No	id